Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM: Design Butterworth and Chebyshev filter using bilinear transformation method.**	
Experiment No: 4	**Date:15-10-2026**	**Enrollment No:92400133121**

Aim: Design Butterworth and Chebyshev filter using bilinear transformation method.

Theory:

- The bilinear transformation method is commonly used to design analog filters and then convert them into digital filters.
- This method maps the analog frequency response to the digital frequency response using a bilinear transformation.
- The Butterworth and Chebyshev filters are two commonly used filter types.
- The Butterworth filter has a maximally flat frequency response in the passband, while the Chebyshev filter allows for a sharper transition between the passband and the stopband at the expense of ripples in either the passband or stopband.

Butterworth Filter

Code:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import butter, bilinear, freqz, cheby1

def bwf(filter_order,cutoff_frequency,sampling_frequency):

    analog_b,analog_a = butter(filter_order,cutoff_frequency,analog=True,btype='low') #continuous signal + low pass filter

    print(analog_b)
    print(analog_a)

    digital_b,digital_a = bilinear(analog_b,analog_a,sampling_frequency) #perform bilinear

    return digital_b,digital_a

filter_order = 4
cutoff_frequency = 1000
sampling_frequency = 8000

digital_b,digital_a = bwf(filter_order,cutoff_frequency,sampling_frequency)

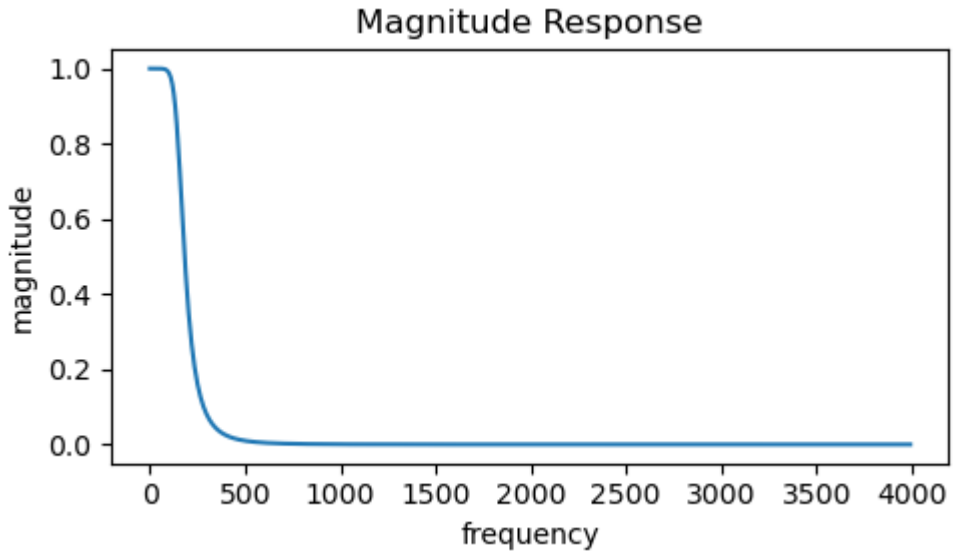
frequency, magnitude_response = freqz(digital_b,digital_a,fs=sampling_frequency)

plt.figure(figsize=(5,3))
plt.plot(frequency,np.abs(magnitude_response))
```

```
plt.title("Magnitude Response")
plt.xlabel("frequency")
plt.ylabel("magnitude")
plt.tight_layout()

plt.show()
```

LOW PASS
ORDER = 4
CUTOFF = 1000

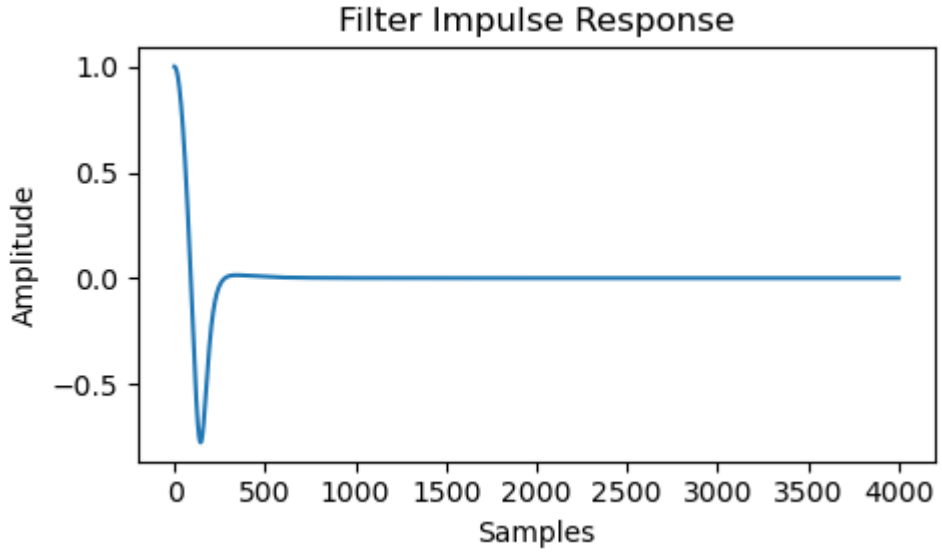



```
frequency,impulse_response = freqz(digital_b,digital_a,fs=sampling_frequency,worN=4000)

plt.figure(figsize=(5,3))
plt.plot(impulse_response)
plt.title("Filter Impulse Response")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.tight_layout()

plt.show()
```

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Chebyshev Filter

Code:

```
def cf(filter_order,cutoff_frequency,sampling_frequency,ripple):

    wc = 2*sampling_frequency*np.tan(np.pi*cutoff_frequency/sampling_frequency)
    analog_b,analog_a = cheby1(filter_order,ripple,wc,analog=True,btype='low') #continuous signal +
low pass filter

    print(analog_b)
    print(analog_a)

    digital_b,digital_a = bilinear(analog_b,analog_a,sampling_frequency) #perform bilinear

    return digital_b,digital_a
filter_order = 4
cutoff_frequency = 1000
sampling_frequency = 8000
ripple = 0.5

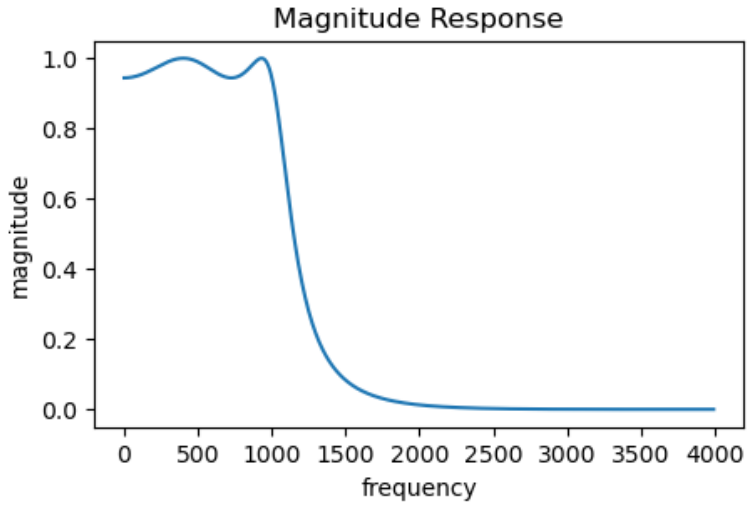
digital_b,digital_a = cf(filter_order,cutoff_frequency,sampling_frequency,ripple)

frequency, magnitude_response = freqz(digital_b,digital_a,fs=sampling_frequency)

plt.figure(figsize=(5,3))
```

```
plt.plot(frequency,np.abs(magnitude_response))
plt.title("Magnitude Response")
plt.xlabel("frequency")
plt.ylabel("magnitude")

plt.show()
```

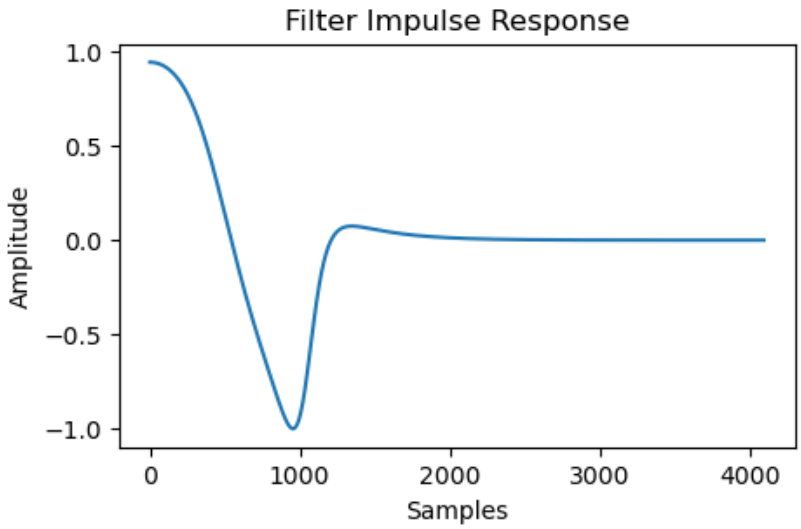



```
frequency,impulse_response = freqz(digital_b,digital_a,fs=sampling_frequency,worN=4096)

plt.figure(figsize=(5,3))
plt.plot(impulse_response)
plt.title("Filter Impulse Response")
plt.xlabel("Samples")
plt.ylabel("Amplitude")

plt.show()
```

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Conclusion

- Butterworth and Chebyshev low-pass digital filters were designed using the bilinear transformation method.
- The analog filter coefficients were calculated and converted into digital filter coefficients using bilinear transformation.
- The magnitude response and impulse response of both filters were plotted and observed.
- The Butterworth filter showed a smooth, maximally flat passband response.
- The Chebyshev filter showed a sharper transition from the passband to the stopband, with a small ripple in the passband.